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Folding Board Game Table

Abstract

Provided is a folding board game table, including a table board, support legs, a hinge seat assembly, and gas springs. The table board is formed by two mutually hinged board bodies. The two support legs are provided and are symmetrically hinged at two ends of a lower surface of the table board. The hinge seat assembly includes first hinge seats fixed on support beams and second hinge seats fixed on bottom surfaces of the board bodies. The gas springs are disposed between the first hinge seats and the second hinge seats. The folding board game table is compact in structure, convenient and labor-saving to fold and unfold, easy to store and carry, good in usage effect, and high in safety.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims the benefit of Chinese Patent Application No. 202420445087.2 filed on Mar. 7, 2024, the contents of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to a chess and card table, and in particular to a folding board game table.

BACKGROUND

[0003] Chess and card tables and other game tables are commonly used furniture in daily life, which are used for multi-player entertainment activities such as chess and cards. However, when not in use, they occupy indoor space and are inconvenient to move.

SUMMARY

[1] Technical Problem to be Solved

[0004] The technical problem to be solved by the present disclosure is to provide a folding board game table with a compact structure, which can be folded, occupies a small space, and is easy to carry.

[2] Technical Solution to the Problem

[0005] The present disclosure provides a folding board game table, including: [0006] a table board **1**, formed by two mutually hinged board bodies; [0007] two support legs **3**, symmetrically hinged at two ends of a lower surface of the table board **1**, where each of the support legs **3** includes a hinged shaft **31** and support rods fixed on the hinged shaft **31**, the hinged shaft **31** is rotatably mounted at a lower end of the table board **1**, and a rotation axis of the hinged shaft is parallel to a rotation axis of the corresponding board body; and two support rods are provided and are symmetrically disposed at both ends of the hinged shaft **31**, and a support beam **33** parallel to the hinged shaft **31** is fixed between the two support rods; [0008] a hinge seat assembly, including first hinge seats **35** fixed on the support beams **33** and second hinge seats **5** fixed on bottom surfaces of the board bodies, the first hinge seats **35** and the second hinge seats **5** being located on a same vertical plane; and [0009] gas springs **4**, each of the gas springs including a cylinder body **41** and a piston rod **42** disposed on the cylinder body **41** in a manner of sliding fit, where the cylinder body **41** is internally filled with inert gas that causes the piston rod **42** to move outward, an end portion of the cylinder body **41** is hinged on the corresponding second hinge seat **5**, and an end portion of the piston rod **42** is hinged on the corresponding first hinge seat **35**.

[0010] Further, when the support legs **3** are in an unfolded state, an included angle between each of the support legs and the table board **1** is 90 degrees; and when in a folded state, the support legs **3** are parallel to the lower surface of the table board **1**.

[0011] Further, the table board **1** is provided with limiting blocks for limiting rotation strokes of the support legs **3**.

[0012] Further, when the support legs **3** are unfolded to limit positions, the piston rods **42** are just at outer limit positions.

[0013] Further, each of the second hinge seats **5** includes a bottom plate **51** mounted on the lower surface of the table board in a fitted manner by bolts, and bracket plates **52** are vertically fixed on the bottom plate **51**; a plane where the bracket plates **52** are located is perpendicular to the rotation axis of the corresponding board body; and the two bracket plates **52** are provided and are disposed in parallel, and a hinge area for accommodating the corresponding cylinder body is formed between the two bracket plates **52**.

[0014] Further, the folding board game table further includes a main shaft **2**, where a rotary connection between the two board bodies is achieved by the main shaft **2**, the second hinge seats **5** are provided with support plates **53**, and shaft holes are formed in the support plates **53** to allow the main shaft **2** to pass through and achieve a hinged connection.

[0015] Further, each of the support rods includes a first rod body **321** rigidly connected to the hinged shaft **31**, an end portion of the first rod body **321** is bent toward one side to form a second rod body **322**, and an end portion of the second rod body **322** is bent reversely to form a third rod body **323** parallel to the first rod body **321**; and the support beam **33** is fixed between the two first rod bodies **321**.

[0016] Further, a reinforcing rod **34** is disposed between the two second rod bodies **322** or between the two third rod bodies **323** on the same support leg.

[0017] Further, a rectangular frame **6** is fixed at a lower end of each of the board bodies, and the support legs **3** are hinged on the rectangular frames **6**.

[0018] Further, the rectangular frames **6** are made of steel pipes.

[3] Beneficial Effects

[0019] The folding board game table provided by the present disclosure adopts the folding table legs, which can be folded when not in use, so that the space occupation is reduced, and the table is easy to store and carry, and has high convenience. A table top is formed by hinge joint with a structure of the double board bodies, which can be folded in half, greatly reducing space occupation and lowering transportation costs. Due to adoption of a gas spring structure, quick unfolding and folding can be achieved, with convenient and labor-saving operation, and the reliability and stability are improved. The structures of the table legs are optimized for achieving placement stability and reliability. A main shaft structure is provided as a hinge center to improve connection stability of the board bodies. The hinge seats are connected to the main shaft, which can enable the main shaft to be subjected to the axial force applied by the gas springs, thus avoiding the thrust on the table, and ensuring that the table is stable and reliable in use, and high in safety. The folding board game table provided by the present disclosure is compact in structure, convenient and labor-saving to fold and unfold, easy to store and carry, good in usage effect, and high in safety.

Description

BRIEF DESCRIPTION OF FIGURES

[0020] FIG. **1** is a front view of a folding board game table according to the present disclosure;

[0021] FIG. **2** is a schematic structural diagram of the folding board game table according to the present disclosure;

[0022] FIG. **3** is a schematic diagram showing the installation of support legs of the folding board game table according to the present disclosure;

[0023] FIG. **4** is a schematic diagram showing the installation of a gas spring of the folding board game table according to the present disclosure;

[0024] FIG. **5** is a schematic structural diagram of the support leg of the folding board game table according to the present disclosure; and

[0025] FIG. **6** is an enlarged view of Part A in FIG. **3**.

DETAILED DESCRIPTION

[0026] The embodiments of the present disclosure will be described in detail below in conjunction with the accompanying drawings.

[0027] Referring to FIG. **1** to FIG. **6**, the present disclosure provides a folding board game table, which includes a table board **1**, support legs **3**, a hinge seat assembly, and gas springs **4**.

[0028] The table board **1** is formed by two mutually hinged board bodies, which can be folded downwards. An upper surface of table board **1** serves as a working surface for chess and card entertainment.

[0029] Two support legs **3** are provided, and are symmetrically hinged at two ends of a lower surface of the table board **1** for supporting the table board **1**, and limiting blocks are disposed on the table board **1** and configured to limit rotation strokes (angles) of the support legs **3** to avoid

overtravel during rotation. Each of the support legs **3** includes a hinged shaft **31** and support rods fixed on the hinged shaft **31**. A length direction of the hinged shaft **31** is parallel to a rotation axis of the corresponding board body **1**, that is, parallel to a folding axis of the corresponding board body. The hinged shaft is rotatably mounted at a lower end of the table board **1**, and a rotation axis of the hinged shaft is parallel to the rotation axis of the corresponding board body. Two support rods are provided and are symmetrically disposed at both ends of the hinged shaft **31**. A support beam **33** is fixed between the two support rods, and the support beam **33** is parallel to the hinged shaft **31**.

[0030] Specifically, each of the support rods includes a first rod body **321**, and the first rod body **321** is rigidly connected to the hinged shaft **31**. An end portion of the first rod body **321** is bent toward one side by 30° to 45° to form a second rod body **322**, and an end portion of the second rod body **322** is bent reversely to form a third rod body **323**. The third rod body **323** is parallel to the first rod body **321**, and the support beam **33** is fixed between the two first rod bodies **321**. The rod bodies on the same entire support leg **3** are located on a same plane. In order to improve the structural strength, a reinforcing rod **34** is disposed between the two second rod bodies **322** or between the two third rod bodies **323** on the same support leg, so as to improve the overall structural strength of the support leg **3**.

[0031] The hinge seat assembly includes first hinge seats **35** and second hinge seats **5**. The first hinge seats **35** are fixed on the support beams **33**, and the second hinge seats **5** are fixed on bottom surfaces of the board bodies. The first hinge seats **35** and the second hinge seats **5** are located on a same vertical plane, specifically on a middle vertical plane of the table board.

[0032] Each of the gas springs **4** includes a cylinder body **41** and a piston rod **42**. The piston rod **42** is disposed on the cylinder body **41** in a manner of sliding fit. The cylinder body **41** is internally filled with inert gas that causes the piston rod **42** to move outward. An end portion of a tail part of the cylinder body **41** is hinged on the corresponding second hinge seat **5**, and an end portion of a head part of the piston rod **42** is hinged on the corresponding first hinge seat **35**. The connection between the support legs and the board bodies is achieved by the gas springs.

[0033] Each of the second hinge seats **5** includes a bottom plate **51**, and the bottom plate **51** is mounted on the lower surface of the table board in a fitted manner by bolts. Bracket plates **52** are vertically fixed on the bottom plate **51**. A plane where the bracket plates **52** are located is perpendicular to the rotation axis of the corresponding board body. The two bracket plates **52** are provided and are disposed in parallel, and a hinge area for accommodating the corresponding cylinder body is formed between the two bracket plates **52**, so as to achieve a hinged connection.

[0034] In the present application, a main shaft **2** is also included. A rotary connection between the two board bodies is achieved by the main shaft **2**. Furthermore, the second hinge seats **5** are provided with support plates **53**, and shaft holes are formed in the support plates **53** to allow the main shaft **2** to pass through and achieve hinged support, which can withstand the thrust generated by the gas springs during unfolding, thus preventing the table board from directly bearing the thrust applied by the gas springs, and improving the overall structural strength and reliability.

Furthermore, a rectangular frame **6** is fixed at a lower end of each of the board bodies, and the support legs **3** are hinged on the rectangular frames **6**. At the same time, the rectangular frames are hinged on the main shaft to form an integral support for the board game table. The rectangular frames **6** are made of steel pipes.

[0035] When the support legs **3** are in an unfolded state, the included angle between each of the support legs and the table board **1** is 90 degrees, that is, the support leg and the table board are mutually perpendicular. When in a folded state, the support legs **3** are parallel to the lower surface of the table board **1**, or directly attached to the lower surface of table board **1**.

[0036] In this embodiment, when the support legs **3** are unfolded to limit positions, the piston rods **42** are just at outer limit positions. At this time, even without the obstruction of the support legs, the piston rods **42** will not continue to extend under the action of internal pressure, that is, the piston

rods are already in the extended limit positions.

[0037] The folding board game table provided by the present disclosure adopts the folding table legs, which can be folded when not in use, so that the space occupation is reduced, and the table is easy to store and carry, and has high convenience. A table top is formed by hinge joint with a structure of the double board bodies, which can be folded in half, greatly reducing space occupation and lowering transportation costs. Due to adoption of a gas spring structure, quick unfolding and folding can be achieved, with convenient and labor-saving operation, and the reliability and stability are improved. The structures of the table legs are optimized for achieving placement stability and reliability. A main shaft structure is provided as a hinge center to improve connection stability of the board bodies. The hinge seats are connected to the main shaft, which can enable the main shaft to be subjected to the axial force applied by the gas springs, thus avoiding the thrust on the table, and ensuring that the table is stable and reliable in use, and high in safety. The folding board game table provided by the present disclosure is compact in structure, convenient and labor-saving to fold and unfold, easy to store and carry, good in usage effect, and high in safety.

[0038] The foregoing descriptions are merely exemplary embodiments of the present disclosure. It should be pointed out that for those of ordinary skill in the art, some improvements and embellishments without departing from the principle of the present disclosure can also be made, which shall also be regarded as the protection scope of the present disclosure.

Claims

1. A folding board game table, comprising: a table board, formed by two mutually hinged board bodies; two support legs, symmetrically hinged at two ends of a lower surface of the table board, wherein each of the support legs comprises a hinged shaft and support rods fixed on the hinged shaft, the hinged shaft is rotatably mounted at a lower end of the table board, and a rotation axis of the hinged shaft is parallel to a rotation axis of the corresponding board body; and two support rods are provided and are symmetrically disposed at both ends of the hinged shaft, and a support beam parallel to the hinged shaft is fixed between the two support rods; a hinge seat assembly, comprising first hinge seats fixed on the support beams and second hinge seats fixed on bottom surfaces of the board bodies, the first hinge seats and the second hinge seats being located on a same vertical plane; and gas springs, each of the gas springs comprising a cylinder body and a piston rod disposed on the cylinder body in a manner of sliding fit, wherein the cylinder body is internally filled with inert gas that causes the piston rod to move outward, an end portion of the cylinder body is hinged on the corresponding second hinge seat, and an end portion of the piston rod is hinged on the corresponding first hinge seat.
2. The folding board game table according to claim 1, wherein when the support legs are in an unfolded state, an included angle between each of the support legs and the table board is 90 degrees; and when in a folded state, the support legs are parallel to the lower surface of the table board.
3. The folding board game table according to claim 1, wherein the table board is provided with limiting blocks for limiting rotation strokes of the support legs.
4. The folding board game table according to claim 1, wherein when the support legs are unfolded to limit positions, the piston rods are just at outer limit positions.
5. The folding board game table according to claim 1, wherein each of the second hinge seats comprises a bottom plate mounted on the lower surface of the table board in a fitted manner by bolts, and bracket plates are vertically fixed on the bottom plate; a plane where the bracket plates are located is perpendicular to the rotation axis of the corresponding board body; and the two bracket plates are provided and are disposed in parallel, and a hinge area for accommodating the corresponding cylinder body is formed between the two bracket plates.
6. The folding board game table according to claim 5, further comprising a main shaft, wherein a

rotary connection between the two board bodies is achieved by the main shaft, the second hinge seats are provided with support plates, and shaft holes are formed in the support plates to allow the main shaft to pass through and achieve a hinged connection.

7. The folding board game table according to claim 1, wherein each of the support rods comprises a first rod body rigidly connected to the hinged shaft, an end portion of the first rod body is bent toward one side to form a second rod body, and an end portion of the second rod body is bent reversely to form a third rod body parallel to the first rod body; and the support beam is fixed between the two first rod bodies.

8. The folding board game table according to claim 7, wherein a reinforcing rod is disposed between the two second rod bodies or between the two third rod bodies on the same support leg.

9. The folding board game table according to claim 1, wherein a rectangular frame is fixed at a lower end of each of the board bodies, and the support legs are hinged on the rectangular frames.

10. The folding board game table according to claim 9, wherein the rectangular frames are made of steel pipes.
